Original-time freezing of source-displayed PAH-v2 readouts

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1. Purpose and scope

Fix any admitted finite open square $G_{h,N}$, with the original anchors, orientations and elementary faces. Write $v = |V|$, $e = |E|$, $p = |P| > 0$. These are combinatorial counts, not physical dimensions or volume. Fix all parent real parameters, including $0 < \epsilon \le 1$, $\beta, \nu > 0$; the signed mass and quartic coefficients may be negative. Fix $R_0 > 0$ and integers $1 \le q_0 \le m_0$. Only r tends to infinity:

$$K_r = \prod_{i=1}^{r+1} p_i, \quad M_s = 2^r, \quad M_\psi = m_0 4^r, \quad R_{\max} = R_0 2^r, \quad Q = q_0 2^r. \tag{1}$$

The operator-approved preregistration is

`strategy/pa-hyp/PAH-v2-RD-001-prereg-v1.json`
`86021a47783189e6e29f86cd24d14e0e1eaf8cfa2d943f6372099912a5571528`

Its source pins incorporate the literal PAH-v1 functional and generator, PAH-v2-r2 tuple counting and signed partial roots, and the adopted Morton/CRT schedule. Every hidden label, valid incidence, duplicate $K = 2$ channel and original Gibbs weight is retained. The displayed generator is

$$(L_r f)(x) = \sum_{a:\, ax \text{ valid}} c_a(x)[f(ax) - f(x)], \quad c_a(x) = m_a(x) e^{-\beta(\mathcal{F}_r(ax) - \mathcal{F}_r(x))/2}. \tag{2}$$

No extra prefactor is added to this displayed formula. A separately instantiated rate multiplier would have to be carried into the constants, not silently normalized to one. The domain is all $\mathbb{C}^{\Omega_r}$, $S_r(t) = e^{tL_r}$, and $\pi_r = Z_r^{-1} e^{-\beta \mathcal{F}_r}$ is the original full counting state. Time is the original external Markov time, not proper, quantum or Lorentzian time.

Use the displayed metric

$$d(z, z') = \sum_V |s - s'| + \sum_V |\psi - \psi'| + \sum_E |U - U'|. \tag{3}$$

Fix any bounded complex readout F with Lipschitz constant $\ell_F < \infty$ independent of r, invariant under gauge and source-preserving anchored cell automorphisms, and put $f_r(x) = F(s(x), \psi(x), U(x))$. This is direct sampling, not the fixed CRT cylinder of R-574. The old cylinder algebra and comparison remain untouched; their failed target is not repaired by restricting it to these functions.

Target. For every such fixed F and every finite $T \ge 0$,

$$\sup_{0 \le t \le T} \| S_r(t) f_r - f_r \|_{L^2(\pi_r)} \longrightarrow 0. \tag{4}$$

Separately, the fixed invariant readout $W = p^{-1} \sum_{b \in P} \operatorname{Re} U_b$, already motivated by the source plaquette term, has a strictly positive cutoff-uniform Gibbs variance. Neither statement constructs a common limiting state, Hilbert space or infinite-volume process.

2. Full-state bounds and the primary proof

The exact charge constraint, not a tail truncation, gives

$$\sum_V |\psi_v| = (R_{\max}/M_\psi)Q = A := R_0 q_0/m_0, \quad \epsilon \le s_v \le 1, \quad |U_e| = 1. \tag{5}$$

Thus $|\psi_v| \le A$ even though $R_{\max} \to \infty$. For $J_e = 2/(s_v + s_w)$ and the face average J_b, both are at most $1/\epsilon$. Taking absolute values term by term in the pinned functional gives $|\mathcal{F}_r| \le B$, where

$$\begin{aligned} b_0 &= \tfrac{1}{2}\lambda_s(1-\epsilon)^2 + \tfrac{1}{2}|m^2|A^2 + \tfrac{1}{4}|\lambda_4|A^4 + \tfrac{1}{6}\eta_6 A^6 + \tfrac{1}{2}gA^2, \\ B &= vb_0 + \tfrac{1}{2}\kappa_s e(1-\epsilon)^2 + 2\kappa_D eA^2/\epsilon + 2\kappa_g p/\epsilon. \end{aligned} \tag{6}$$

Indeed, $|\psi_w - U_e\psi_v| \le 2A$ and $0 \le 1 - \operatorname{Re} U_b \le 2$. All original mobilities lie in $(0, 1]$; even aperture moves use two admissible endpoints. Therefore

$$c_a(x) \le C := e^{\beta B}. \tag{7}$$

This bound includes negative allowed m^2, λ_4 and every full state. It need not be uniform as geometry grows or $\epsilon \downarrow 0$.

Set $\delta_r = (R_0/m_0)2^{-r}$, $a_r = (1-\epsilon)2^{-r}$ and $k_r = 2\sin(\pi/K_r) \le 2\pi/K_r$. The four signed families have the following upper bounds in (3):

family	number of labels	$d(z(ax), z(x))$
PH	$2v$	Ak_r
TR	$2e$	$2\delta_r$
LK	$2e$	k_r
AP	$2v$	a_r

(8)

For TR the occupation changes but both phase coordinates remain fixed; the two complex increments each have magnitude δ_r. For PH/LK use chord distance across modular wrap. Invalid roots are absent, so counting all labels is an upper bound, not an endpoint modification. Equations (2), (7), (8) yield

$$\begin{aligned} \|L_r f_r\|_\infty &\le \ell_F C D_r, \\ D_r &= [2v(1-\epsilon) + 4eR_0/m_0]2^{-r} + 4\pi(vA + e)/K_r. \end{aligned} \tag{9}$$

For completeness the finite semigroup step requires no invariant smooth subspace. Let $q \ge \max_x \sum_{y \ne x} L_r(x, y)$, taking $q > 0$. Then $P = I + L_r/q$ is stochastic and $S_r(t) = e^{-qt}\sum_{n \ge 0}(qt)^n P^n/n!$ is a sup-norm contraction on the full finite space. Differentiate the matrix exponential to obtain

$$S_r(t)f_r - f_r = \int_0^t S_r(u)L_r f_r \, du, \qquad \sup_{t \le T} \|S_r(t)f_r - f_r\|_{L^2(\pi_r)} \le T\ell_F C D_r. \tag{10}$$

Normalized π_r gives the last norm comparison. Since $K_r \ge 2^{r+1}$, $D_r \to 0$ geometrically. This proves (4) for every fixed F, not uniformly over unbounded Lipschitz constants. It neither identifies $S_r(t)f_r$ as a sampled Lipschitz function nor compares different r spaces.

3. Independent reversible-state derivation

An independent derivation avoids the pointwise rate cap. Each fixed signed root is a partial bijection with inverse root. The original midpoint rate and symmetric mobility give exactly

$$\pi_r(x)c_a(x) = m_a(x)\sqrt{\pi_r(x)\pi_r(ax)}, \qquad \sum_{x:\, ax \text{ valid}} \pi_r(x)c_a(x) \le 1. \tag{11}$$

The inequality follows from Cauchy–Schwarz on the valid domain and its injective image, each of probability mass at most one, and $m_a \le 1$. It remains true for omitted endpoints and distinct labels with the same map. Pairing inverse incidences gives the exact full Gibbs Dirichlet form

$$\mathcal{E}_r(f) = \langle f, -L_r f\rangle_{\pi_r} = \tfrac{1}{2} \sum_{x,a \text{ valid}} \pi_r(x) c_a(x) |f(ax) - f(x)|^2. \tag{12}$$

Thus (8), (11) imply

$$\mathcal{E}_r(f_r) \le \ell_F^2 [vA^2 k_r^2 + 4e\delta_r^2 + ek_r^2 + va_r^2] \le \ell_F^2 H 4^{-r}, \tag{13}$$

where $H = \pi^2(vA^2 + e) + 4e(R_0/m_0)^2 + v(1-\epsilon)^2$. The finite matrix $-L_r$ is self-adjoint nonnegative in $L^2(\pi_r)$ by (12). Diagonalize this finite matrix; for each eigenvalue $\lambda \ge 0$, $(1 - e^{-t\lambda})^2 \le t\lambda$. For $x = t\lambda \le 1$ use $1 - e^{-x} \le x$, and for $x \ge 1$ use $1 - e^{-x} \le 1$. Summing the squared spectral coefficients proves independently

$$\sup_{t \le T} \|S_r(t) f_r - f_r\|_{L^2(\pi_r)} \le \ell_F \sqrt{TH}\, 2^{-r}. \tag{14}$$

This is finite linear algebra at each cutoff, not an infinite-dimensional spectral theorem. Neither irreducibility nor a spectral gap is assumed.

4. Positive variance: not a constant-observable artifact

The source face readout W is bounded and real. Closed holonomies are gauge invariant; real parts are unchanged by orientation reversal, and averaging over all elementary faces is invariant under every actual source automorphism. Telescoping products of unit links give $|W(U) - W(U')| \le (2/p) \sum_e |U_e - U'_e|$, since each edge lies in at most two faces. In particular W is one fixed admissible Lipschitz readout.

Let γ_r be normalized uniform counting on the full tuple space, used only as a comparison measure. The fixed-Q constraint acts on occupations only; link coordinates under γ_r are independent uniform $\mathbb{Z}_{K_r}$. For every elementary face its signed link character has zero mean. Distinct elementary faces possess an edge in their symmetric difference with coefficient 1 or -1, so the character averages in both $U_b U_c$ and $U_b \overline{U_c}$ vanish. This proves zero covariance; it does not assert independence of adjacent face variables. Also $\mathbb{E}_{\gamma_r} (\operatorname{Re} U_b)^2 = 1/2$ for $K_r \ge 3$. Consequently

$$\mathbb{E}_{\gamma_r} W = 0, \qquad \operatorname{Var}_{\gamma_r}(W) = 1/(2p) \quad (r \ge 1). \tag{15}$$

At $K_0 = 2$ the variance is $1/p$, so the lower bound below remains safe. From (6), without assuming $\mathcal{F}_r \ge 0$, $d\pi_r / d\gamma_r = e^{-\beta \mathcal{F}_r} / \mathbb{E}_{\gamma_r} e^{-\beta \mathcal{F}_r} \ge e^{-2\beta B}$. Minimization over the centering constant, not a false claim that the Gibbs mean is zero, now gives

$$\operatorname{Var}_{\pi_r}(W) = \inf_{a \in \mathbb{R}} \mathbb{E}_{\pi_r} (W - a)^2 \ge e^{-2\beta B} \inf_a \mathbb{E}_{\gamma_r} (W - a)^2 \ge \frac{e^{-2\beta B}}{2p} > 0. \tag{16}$$

Equations (14) and (16) distinguish vanishing dynamics from vanishing static variance. The lower bound is not claimed uniform in h, N, ϵ, β.

5. Applicability and verification boundary

Norris, *Markov Chains*, chapter 2, sections 2.1 and 2.10, is primary background for finite Q-matrices and exponentials (author's Cambridge index and publisher chapter are located in the preregistration). No inaccessible theorem is imported: the contraction and integral identity are derived above. The only additional standard tool in the independent proof is diagonalization of a finite Hermitian matrix. Its hypotheses follow explicitly from (11)–(12). The new model-specific work is (5)–(9), (11)–(16), not a novelty claim about finite Markov theory.

R-570's source finite domain, inverse labels, balance and symmetry conclusions apply to exactly the pinned v2 revision; (11)–(12) also rederive the needed balance bridge. R-511 instead fixes $K = 2, M_s = 1$ and removes only the radial mesh, so it does not supply this all-cutoff theorem. R-573/R-574 remain negative controls: their fixed CRT character has cutoff-dependent displayed frequency and is not a single fixed readout with a uniform Lipschitz constant. No source-owner, common-core or physical-sector assumption is discharged by changing that distinction.

Primary symbolic/source-root tests, a non-importing independent full finite generator corroboration, and hostile controls have separate scripts and run JSONs. Finite tests check implementations, not the quantifiers proved here. The integrated reproduction command is

```
python -X utf8 verification/scripts/pah_v2_rd001_verify.py
```

The pinned Lean entrypoint checks only the parameterized finite-sum, centering and positive-variance bridges listed in its run receipt. Full source geometry, functional estimates, Fourier orthogonality, finite semigroup identities and the analytic limit are not fully Lean-encoded. Independent implementations/review are in-session, not external-person certification; external adversarial review is invited.

6. Devil's-advocate review

1. **Unbounded amplitudes invalidate the cap. DISMISSED within scope:** (5) follows from the full fixed-Q_r constraint; no discarded tail or replacement state. Different charge scaling requires new analysis.

2. **Negative quadratic/quartic terms. DISMISSED:** their absolute coefficients occur in (6); neither energy positivity nor convexity is used.

3. **Missing sign or half factor. DISMISSED:** the rate cap is $e^{\beta B}$, density floor $e^{-2\beta B}$, and (12) has exactly one half for directed incidences. Mutation controls test these distinctions.

4. **Principal-angle wrap and duplicate roots. DISMISSED:** (3) is complex chord distance; both $K = 2$ signed labels remain. Invalid endpoints are omitted, not clipped or reflected.

5. **A constant readout fakes freezing. DISMISSED:** the fixed source-native W has the uniform positive variance (16). Per-r positivity alone was not used.

6. **Adjacent-face independence is not justified. VALID WITH MITIGATION:** (15) uses only explicit character orthogonality on a distinguishing edge. The Gibbs mean is never set to zero.

7. **The smooth class need not be semigroup invariant. DISMISSED:** (10) and (14) use the full finite semigroup. Only its initial test vector has the stipulated Lipschitz property.

8. **This repairs R-574 or proves physical time. UPHELD as a prohibition:** it does neither. The diagnostic is separate direct sampling; no common limit state, volume-uniform estimate or physical identity follows.

9. **Added rate scale hidden as one. DISMISSED for (2):** the proof uses the displayed literal generator. Any separately instantiated multiplier must appear in its cap and form bound; none is added or fit here.

7. Result footer

Result ID: R-575; PAH-v2-RD-001; exploration EXP-001831.

Precise statement: Equations (10), (14) prove original-time displayed-readout freezing; (16) gives a cutoff-uniform positive invariant plaquette variance.
Scope: Every fixed admitted finite open square, fixed positive epsilon and parent parameters; every fixed preregistered Lipschitz invariant readout; the local-state cutoff alone tends to infinity.
Dependencies: Pinned v1 functional/rates, v2-r2 full tuple/signed roots, adopted cutoff schedule; finite matrix algebra and explicit character sums.
Evidence grade: ANALYTIC, EXACT, EXECUTED; six partial Lean bridges only.
Reproduction command: Assemble the following command on one line.

```
python -X utf8 verification/scripts/pah_v2_rd001_verify.py
  --lean-cache E:/Dev/TECT/verification/lean/.lake/packages
  --elan-home C:/Users/NaEun/.elan
```

Expected output: Primary, independent, hostile, partial Lean and integrated PASS; the preregistered freezing proposition is PROVED.
Falsification gate: An exact source/charge/root/rate, finite contraction, Dirichlet, Fourier or variance-domination error invalidates the proof.
Tier before / after: C6 remains T1 OPEN; the T-054 active gate is unchanged.
No-overclaim statement: No CRT repair, common limiting state/process, physical Pre-A, spacetime, QFT, gravity, horizon, continuum, mass gap or TOE.
Next required action: Stop automatic successors; reopen only for an exact source/proof error or separately approved load-bearing model/physical input. No automatic rate/time rescaling, state change, satellite search or enquiry.